

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (VDE) (CBCS) Degree Examinations
 January 2023

22VDE102 - EMBEDDED SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

Marks BT* CO* PO*

1. a) What is meant by high level noise immunity and low-level noise immunity? Explain variation of logic signal levels on device inputs and outputs with example.
- b) What is dynamic hazard? Explain dynamic hazard with digital circuit and Verilog module.

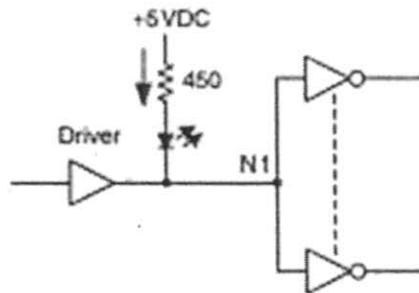
10 L*2 1 3

10 L3 1 3

2. a) Explain Fan-in and Fan-out.

In the circuit shown below, the device identified as the Driver has an output that is connected to a light-emitting diode (LED) circuit and is also fanning out to a number of other gates. When the output of the device is a logical 0, the warning LED is to be illuminated, Driver leading to current flow as shown by the arrow in figure. When the output of the Driver gate is a logical 1, the warning LED is off and no current flows through the LED.

The devices have the following specifications: Inverter $I_{IL} = -400 \mu A$, $I_{IH} = 20 \mu A$, Driver $I_{OL} = 24 \text{ mA}$ for $V_{OL} = 0.2 \text{ V}$, $I_{OH} = -15 \text{ mA}$ for $V_{OH} = 3.5 \text{ V}$, LED current 10 mA. Compute the fan-out for the driving device.



10 L3 1 3

- b) Explain the following:
 - i. Propagation Delay
 - ii. Transport delay
 - iii. Inertial Delay
 - iv. Rise Time
 - v. Fall Time

10 L2 1 3

Unit – II

3. a) What is dynamic modeling with UML? List the important elements of a dynamic model. Explain the different interaction diagrams used in modeling the dynamic behavior of the system.
- b) What are the principal components of UML sequence diagram? Explain Sequence Diagram for making a Time Interval Measurement with a neat diagram.

10 L2 2 3

10 L2 2 3

P.T.O.

4. a) Explain the following.
- UML activity diagram
 - Branch and merge
- b) With necessary UML class diagrams, explain inheritance and interface in UML.
- | | | | |
|----|----|---|---|
| 10 | L2 | 2 | 3 |
| 10 | L2 | 2 | 3 |

Unit – III

5. a) What is a product life cycle model? What are the four life cycle models discussed? Explain briefly the steps that comprise any one of the life cycle models with diagram.
- b) Briefly explain the Co-design process with necessary diagrams.
6. a) Explain the different types of specifications based on the different perspectives in the system with a diagram.
- b) With the neat block diagram explain the hardware architecture of the counter, also explain the data and control flow diagram for the counter System.
- | | | | |
|----|----|---|---|
| 10 | L2 | 3 | 3 |
| 10 | L2 | 3 | 3 |
| 10 | L2 | 3 | 3 |
| 10 | L2 | 3 | 3 |

Unit – IV

7. a) When formulating a debugging strategy, what are some of the major areas that one should consider?
- b) What are the three types of tests generally recognized when debugging and testing modules? Explain with example.
8. a) What is path sensitizing? Explain testing and debugging process, considering basic combinational logic circuit for single variable–single path.
- b) What is boundary-scan testing? Explain the testing strategy with necessary circuit and interface configurations.
- | | | | |
|----|----|---|---|
| 10 | L2 | 4 | 3 |
| 10 | L2 | 4 | 3 |
| 10 | L3 | 4 | 3 |
| 10 | L2 | 4 | 3 |

Unit – V

9. a) What is an operating system? Where it is used and what are its primary functions? With neat sketches explain the basic components of an operating system and their interfaces.
- b) What is kernel? Classify and explain the kernels based on its design.
10. a) List and explain the two types of task scheduling interfaces supported by VxWorks kernel. Also explain the different types of scheduling policies supported by VxWorks.
- b) Explain the following.
- Inter task communication under VxWorks
 - Mutual exclusion mechanisms supported by VxWorks.
- | | | | |
|----|----|---|---|
| 10 | L2 | 5 | 3 |
| 10 | L2 | 5 | 3 |
| 10 | L2 | 5 | 3 |
| 10 | L2 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
